Seoyoung Cho

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Education

Seoul National University, Department of Biosystems and Biomaterials Science and Engineering. Undergraduate student majoring in Biosystems Engineering, Mar. 2023 - Present. GPA: 4.04 / 4.3.

Seoul National University, Interdisciplinary Program in Artificial Intelligence. Double major in Artificial Intelligence, Mar. 2025 - Present.

Research Interests

Vision-Language-Action models.

Reinforcement learning for robotics.

Robot manipulation.

Embodied AI.

Research Experience

Research Intern, Dynamic Robot Control and Design Lab, KAIST. Worked on reinforcement learning-based quadruped locomotion and manipulation, focusing on observation design, termination conditions, and reward design. Jun. 2025 - Aug. 2025.

Projects

Agricultural Robotics Competition: Built an agricultural robot system for orchard environments, including LiDAR-based autonomous navigation, YOLO-based fruit detection, HSV-based line tracking, obstacle detection with a depth camera, and ArUco marker-based return positioning. Sep. 2024 - Nov. 2024.

Two-Speed Gearbox Design: Designed a two-speed gearbox as an integrated transmission system, considering gears, shafts, and the clutch together while selecting materials and dimensions. Used a genetic algorithm to find design parameters that satisfy mechanical constraints, and also considered thermal behavior in the gearbox design process. Mar. 2025 - Jun. 2025.

Humanoid Challenge: Working on the manipulation team for a humanoid robotics challenge, developing a grasping pipeline using point cloud data, GPD-based grasp pose estimation, MoveIt-based motion planning, inverse kinematics, and collision-aware arm movement. Apr. 2026 - Present.

Skills and Techniques

Reinforcement learning for robotics.

Robot manipulation.

ROS.

Programming languages: Python, C++, Java.

Academic Strength: physical system knowledge from Biosystems Engineering combined with programming and mathematical foundations from Artificial Intelligence.